TP53 MUTATIONS THAT ARE ASSOCIATED WITH AN INCREASED COPY NUMBER VARIATION

Cancer Bioinformatics - Final Project

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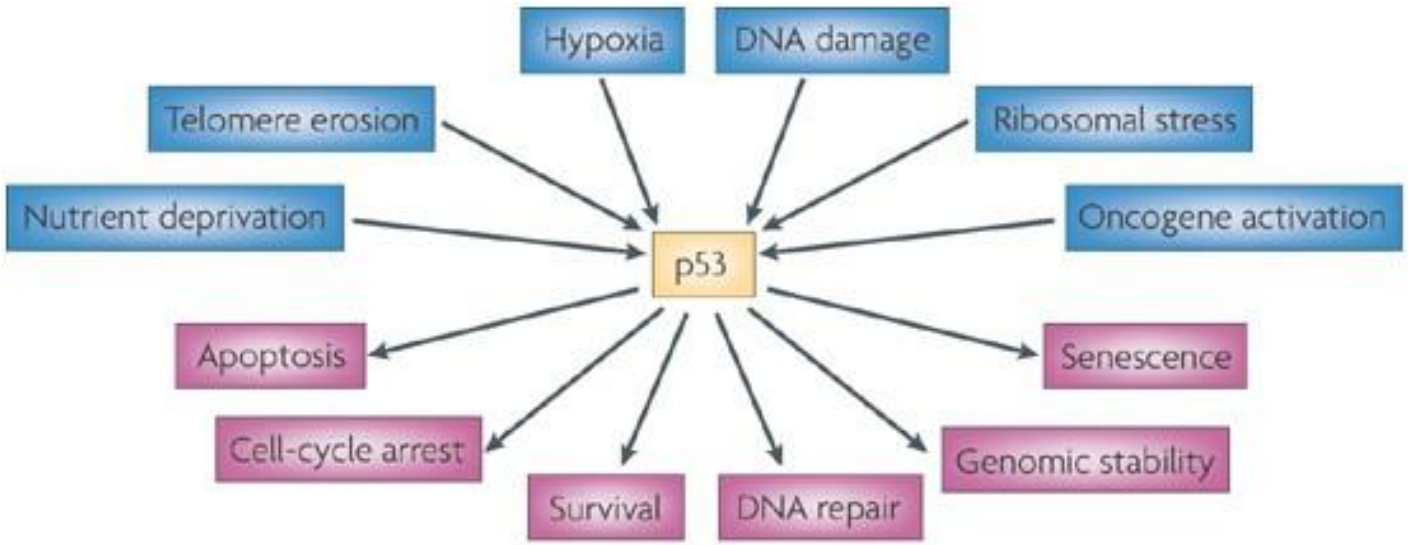
March 19, 2026

TP53

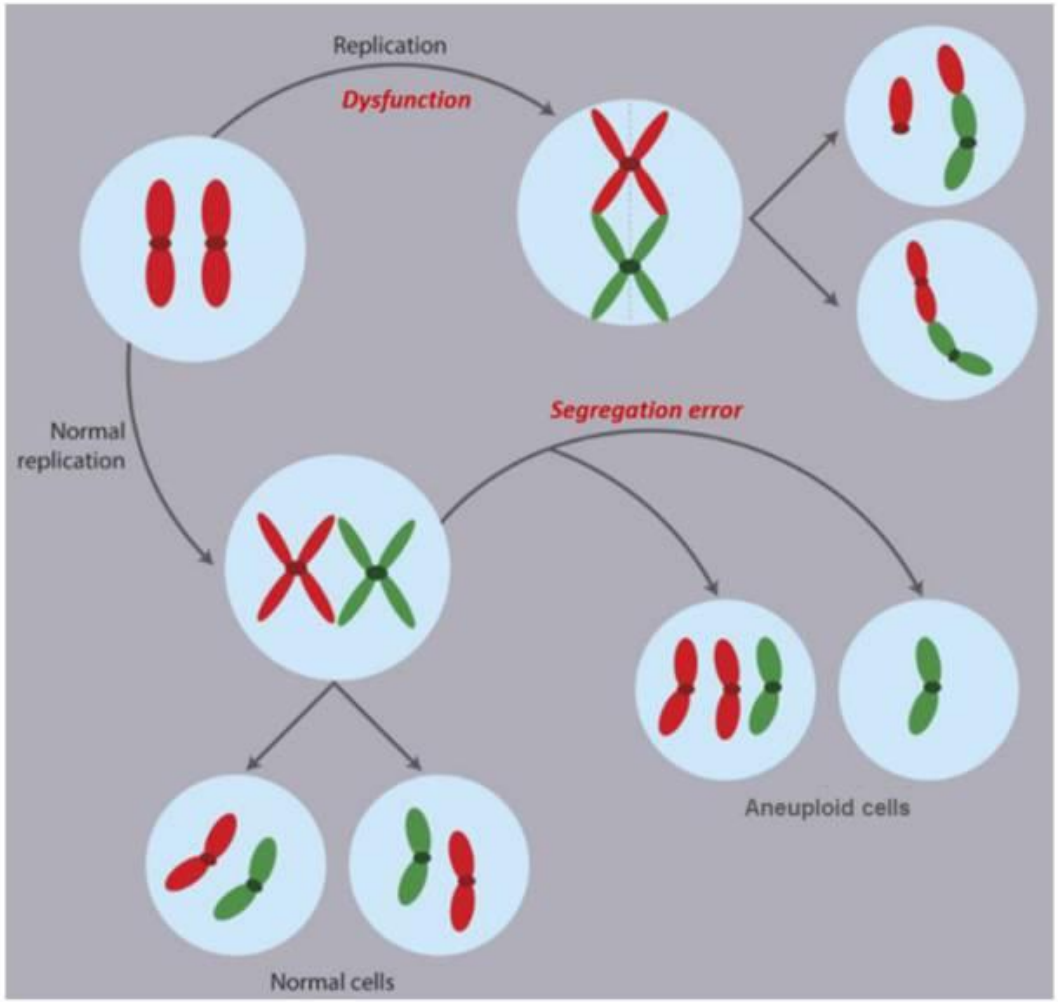


- TP53 acts as the cellular "superhero" responsible for maintaining genomic integrity. It functions as a stress sensor that responds to DNA damage.
- When TP53 is functional, it induces DNA repairs or apoptosis. When broken, errors accumulate.
- TP53 mutations are not binary, and they exist on a spectrum of phenotypic impact.

COPY NUMBER VARIATION

- A stretch of DNA where the number of copies differs from the normal two copies.
  - ⬆ copies → ⬆ RNA → ⬆ protein
  - ⬇ copies → ⬇ protein
  - Example: Oncogene amplification → uncontrolled growth

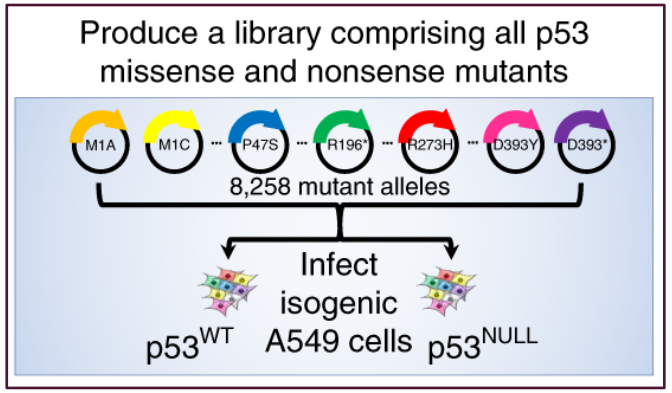
Fraction Genome Altered (FGA) = $\frac{\text{(number of base pairs with copy number gain or loss)}}{\text{(total base pairs profiled)}}$



EXPERIMENTAL FUNCTIONAL SCORE

Screen 8,258 TP53 Variants

1



Combined Phenotype Score (EFS)
Low = Loss of Function
High = WT-like Activity

Quantify Allele Enrichment

2

Calculate Functional Score

3

Giacomelli, A.O., Yang, X., Lintner, R.E. *et al.* Mutational processes shape the landscape of *TP53* mutations in human cancer. *Nat Genet* 50, 1381–1387 (2018).

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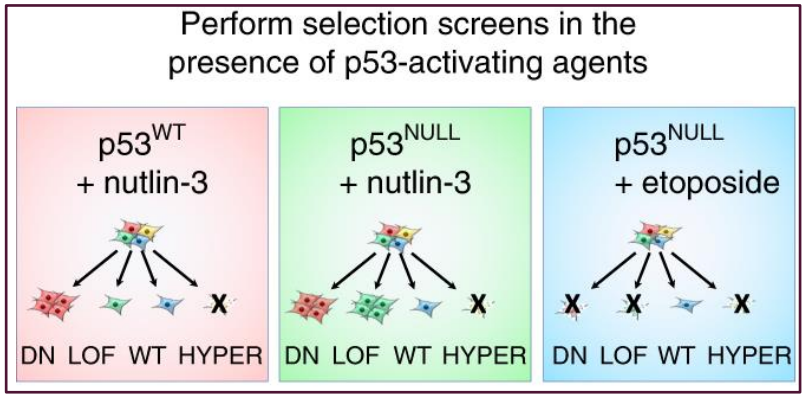

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Screen 8,258 TP53 Variants

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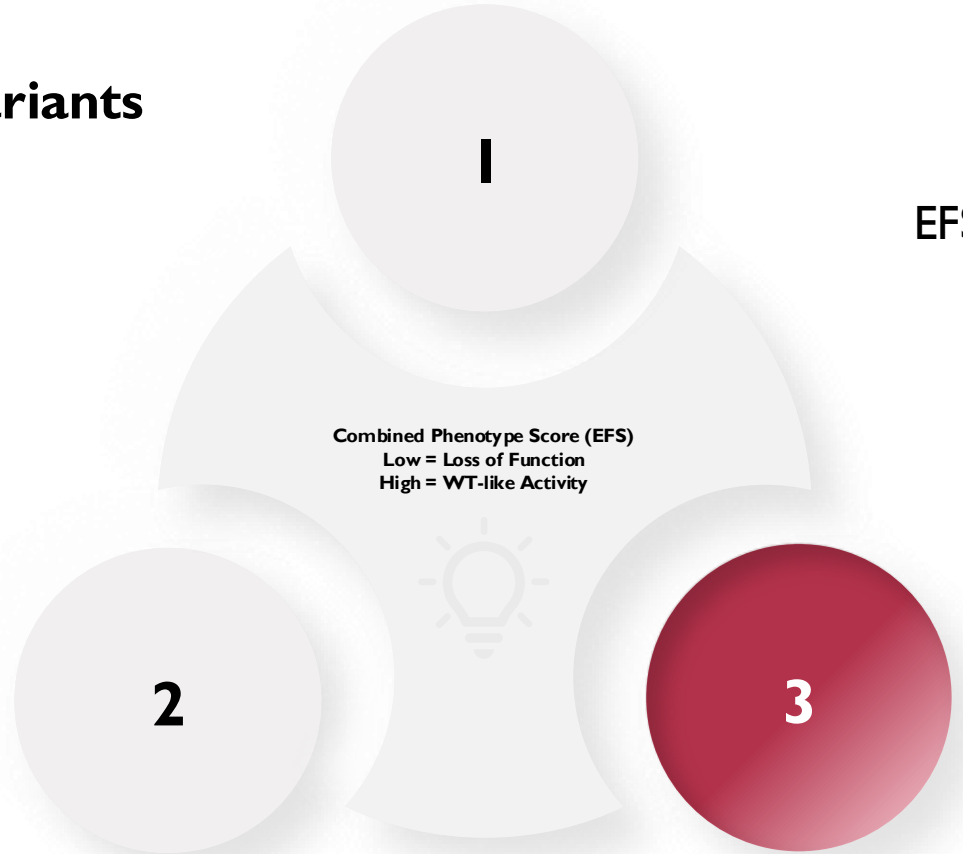
$$EFS = \frac{[Z(WT+nutlin) + Z(NULL+nutlin) - Z(NULL+etop)]}{3}$$



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INITIAL HYPOTHESIS

Patients with TP53 mutations that have the lowest experimental function scores (EFS) will show the highest Fraction Genome Altered (FGA)

RESEARCH QUESTIONS

Is there a correlation between the experimental TP53 EFS (Giacomelli et al.) and patient FGA (GENIE)?

Do "Hotspot" mutations drive higher genomic instability than rare mutations?

Does this correlation hold true across all cancer types or is it specific to certain tissues?

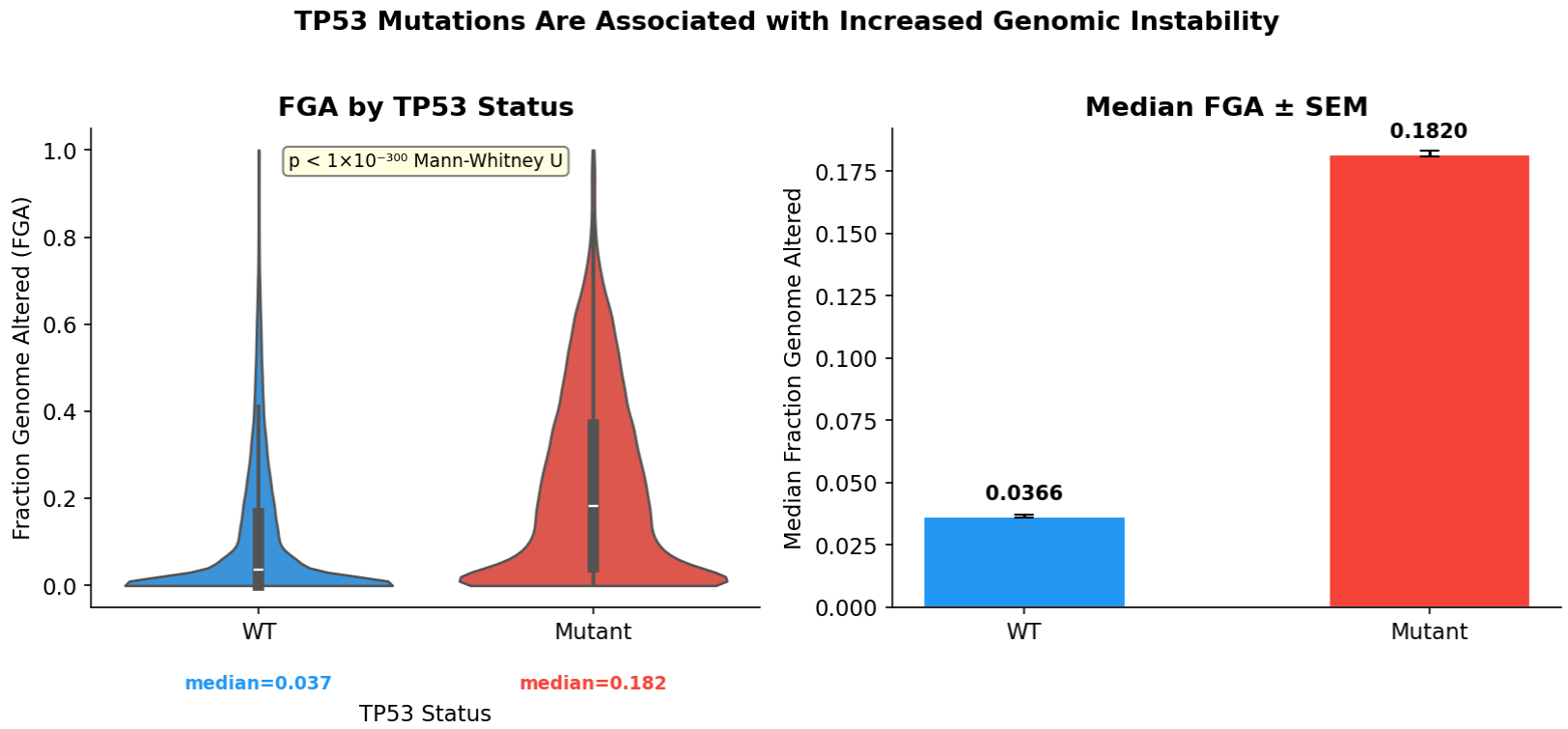
DATA SOURCES: INTEGRATING PHENOTYPE & GENOTYPE

- AACR GENIE v19.0 (~271,000 samples, 100,418 with FGA).
- Role: Real-world patient data - the phenotype
- Metric: FRACTION_GENOME_ALTERED (FGA), a quantitative measure of chromosomal instability
- 35,357 TP53-mutant
- 65,061 TP53-WT

- Giacomelli et al., Nature Genetics, 2018 (25,217 variants, 8,274 unique)
- Role: Experimental "fitness" scores for every possible TP53 mutation - the genotype
- Metric: Experimental Functional Score (derived from cell line fitness screens) quantifies exactly how well each specific TP53 mutant can suppress tumor growth in a controlled lab setting

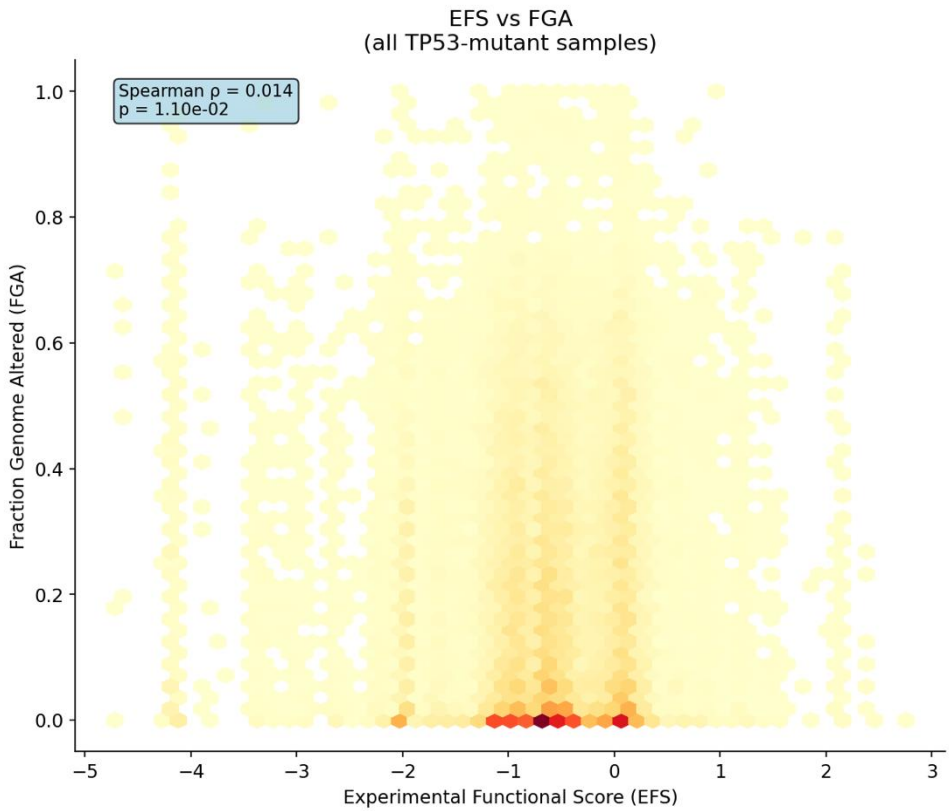
Goal: to show that the physical structure of the genome is breaking down

DO TP53 MUTATIONS CORRELATE WITH HIGHER GENOMIC INSTABILITY (FGA)?



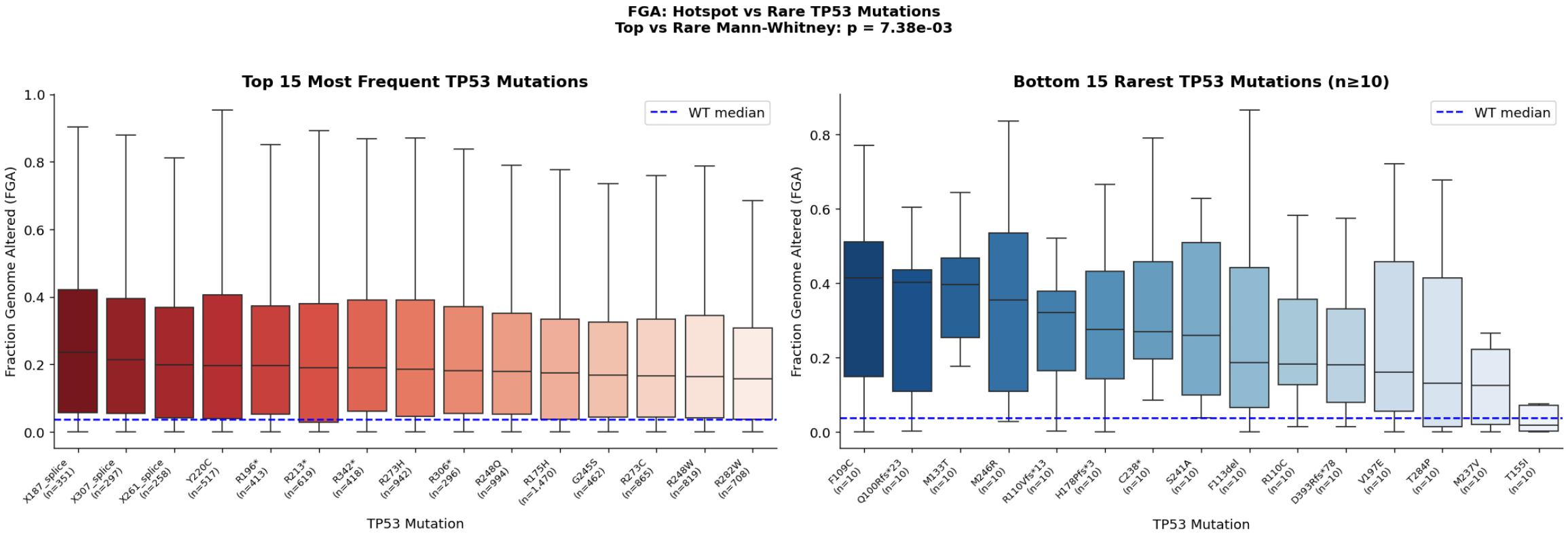
TP53 mutant tumors have significantly higher FGA than WT (median 0.182 vs 0.037, $p < 1\times10^{-300}$)

IS THERE A CORRELATION BETWEEN EFS AND FGA?



Weak correlation. The hypothesis partially holds but EFS alone doesn't predict FGA well

DO "HOTSPOT" MUTATIONS DRIVE HIGHER GENOMIC INSTABILITY THAN RARE MUTATIONS?



Rare mutations have higher median FGA. ”Hotspot” mutations have lower FGA median than rare mutations

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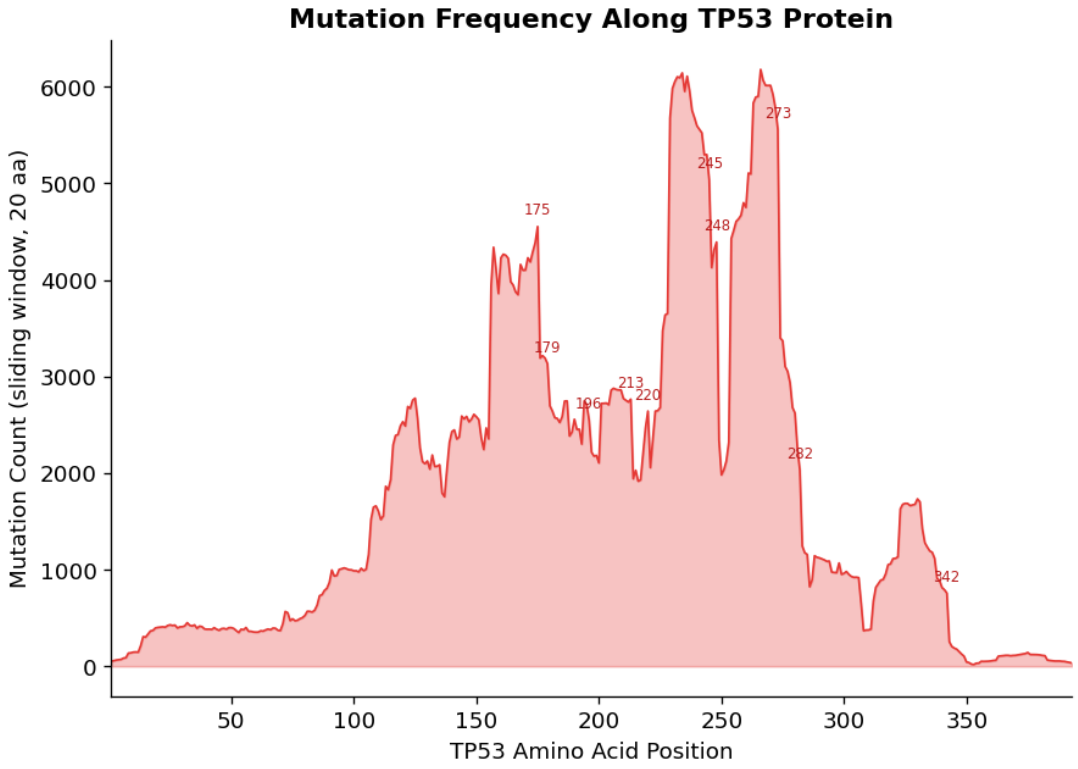
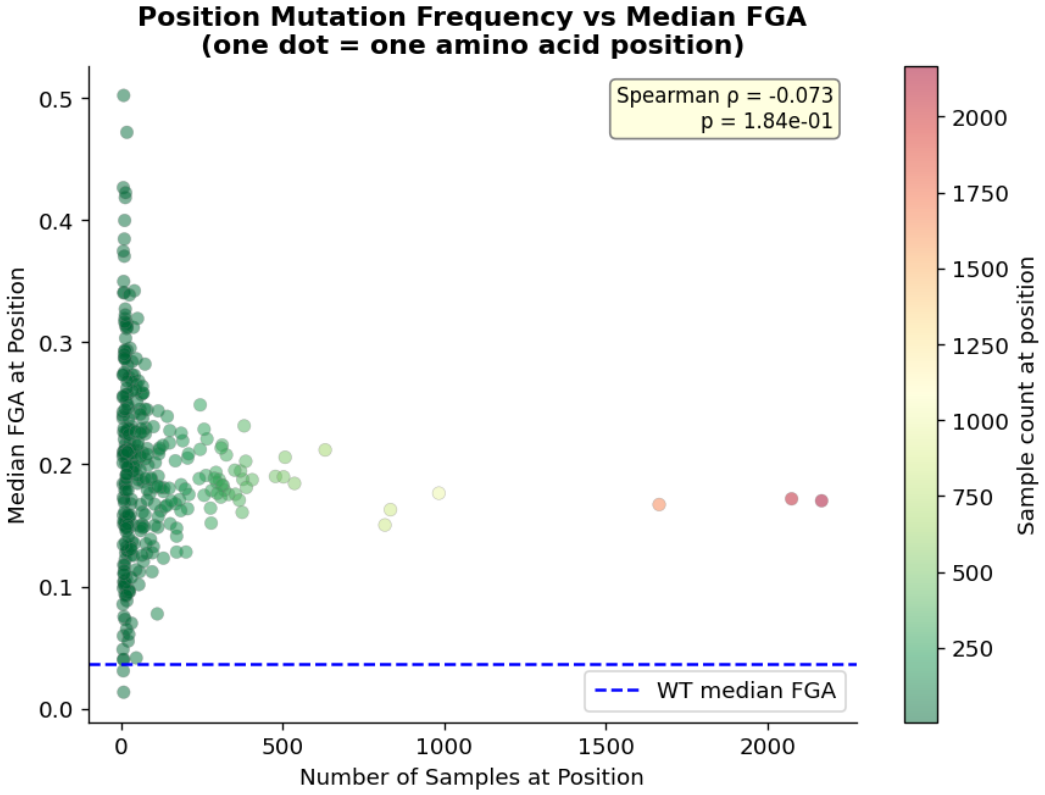
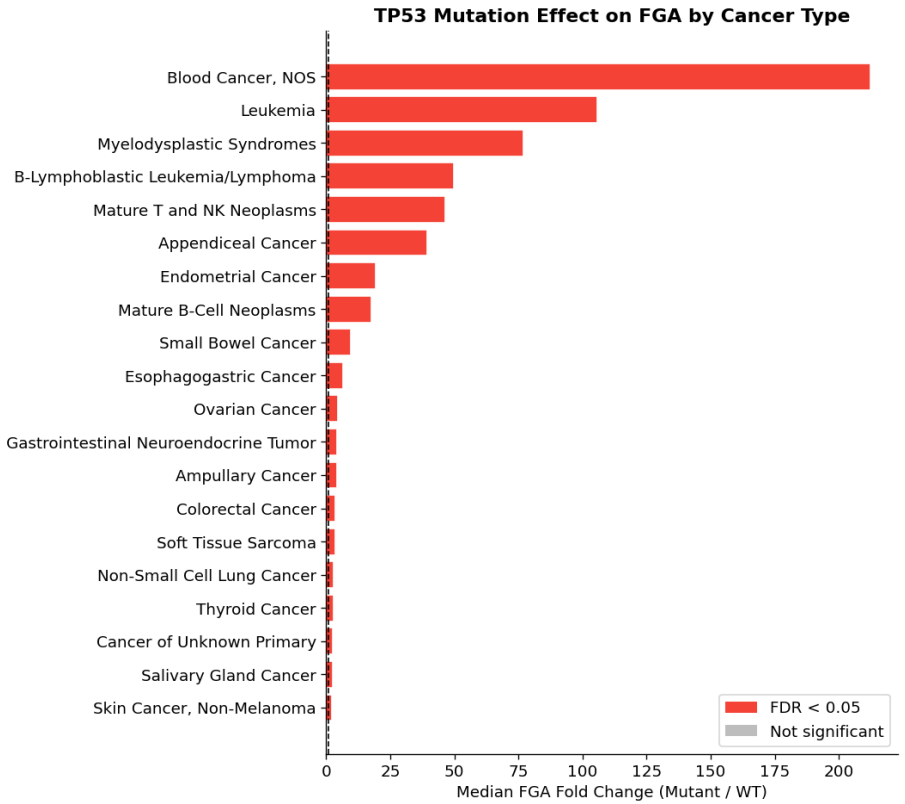
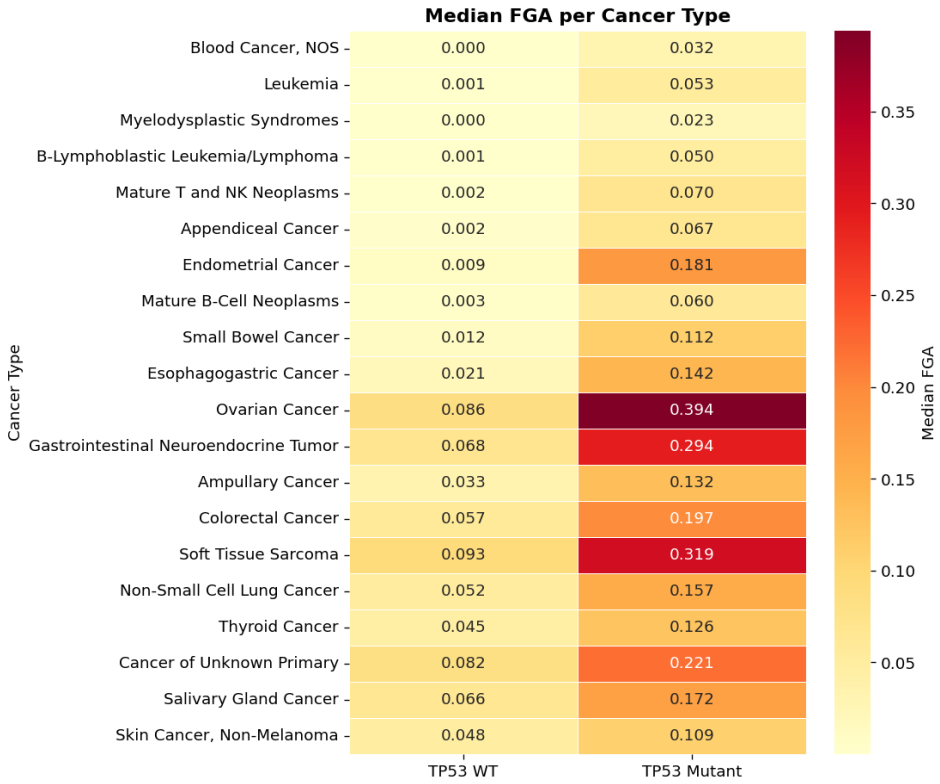




The loss of TP53 function itself drives genomic instability uniformly.

IS THIS CANCER-TYPE SPECIFIC?





All top absolute FGA cancers are solid tumors (Ovarian Cancer, Soft Tissue Sarcoma, and Colorectal Cancer). The blood cancers have the highest fold change but tiny absolute FGA values (~0.03–0.07) because their WT baseline is essentially zero.

CONCLUSIONS

- The binary effect (mutant vs WT) is stronger than the continuous effect (EFS vs FGA).
- This suggests a threshold model: once TP53 changes its function, genomic chaos follows, but the specific degree of functional loss matters less than its presence
- Other tumor-intrinsic and tissue-specific factors play a larger role

CONCLUSIONS

- Hotspot mutations dominate in tumors not because they're functionally the worst (Giacomelli showed this), but because their DNA sequences are highly mutable. Yet they still drive genomic instability, consistent with the threshold model.

CONSTRAINS AND FUTURE DIRECTIONS

- EFS derived from one lung cell line. That may not generalize across tissues
- FGA is a bulk measure, and it doesn't capture which specific chromosomal regions break

THANK YOU!